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Some parts have the sol given

$$S = k \ln \Omega \quad (1)$$

'Entropy decreasing' this doesn't make any physical sense.

3 problems

much easier than prev year, not similar to prev exam,

topics

HARMONIC OSCILLATOR

INDISTINGUISHABLE FREE PARTICLES

MAGNETIC FIELD OR ISING MODEL OR BOTH

these results are LESS PRESENT

Thermodynamic limit, thermodynamic results from ensembles are equivalent.

Memorise

Micro canonical

$$S = k \ln \Omega \quad (2)$$

Canonical

$$\mathcal{F} = -kT \ln Z \quad (3)$$

Grand canonical

$$J = -kT \ln \Xi \quad (4)$$

the parameters they depend on, pressure from $\mathcal{F}$ minus signs.

Discrete system write in words Ω number of micro state configurations leading to given macro state values.

Discrete

$$Z = \sum_n e^{-\beta \epsilon_n} \quad (5)$$

Single particle density of states in **multiple dimensions** and rel or not relativistic

Exact

$$W(E) = \sum_{\epsilon_i} \delta(\epsilon - \epsilon_i) \quad (6)$$

ALWAYS START FROM this as it comes from the phase space then get expression in terms of ϵ or ω using $p = \hbar k$,
 $\omega = \omega(k)$

$$\omega(p) dp = g(n) \frac{V}{h^3} \mathcal{S}_{d-1} p^{d-1} dp \quad (7)$$

p is modulus of p . = in terms of ω or ϵ depends on q asks for.

$$\epsilon = \alpha p^m \quad (8)$$

$$p = \left(\frac{\epsilon}{\alpha}\right)^{1/m} \quad (9)$$

frequency with wave number

Types of systems covered in class

Harmonic oscillator

with all same frequency

and different frequency $\rightarrow$ sound waves propagating in a solid medium. **HARMONIC IS VERY IMPORTANT FOR EXAMS**

Indistinguishable free particles fermions/bosons

bosons to black body, harmonic oscillator should give same result for harmonic oscillator for black body

this isnt as important Fermions fully localised in a magnetic (presence of neg temps)
related to Ising model: thermodynamics phase transitions and correlation functions.

what to know

know how to compute

$$Z = \frac{1}{\left[\cosh\left(\frac{\beta\hbar\omega}{2}\right)\right]^N} \quad (10)$$

$$\Omega = ? \quad (11)$$

we had a problem with

$$\epsilon_n = \hbar\omega \left(n + \frac{1}{2}\right) \quad (12)$$

when do we expect to see quantum results and classical, take limits especially in the harmonic oscillator

$$\mathcal{F} \rightarrow \langle E \rangle \rightarrow C_V \rightarrow \langle n \rangle \quad (13)$$

Taylor expand for small or large temperatures, when classical and when quantum effects. **KNOW THIS** classical result especially, equipartition

$$E = \frac{dNkT}{2} \quad (14)$$

what limit to take to regain the classical expression which is constant. this was the case where frequencies are the same

For different frequencies we use the Debye model

How to determine the debye frequency

$\langle E \rangle$, C_v

taylor expand and limiting behavior

PHOTO GRAPH how does bosons/fermions behave in 3d and other dimensions

INDISTINGUISHABLE PARTICLES

$$\Xi = \prod_i^{\infty} \zeta_i \quad (15)$$

$$\zeta_i = \sum_{n_i} e^{-\beta n_i (\epsilon_i - \mu)} \quad (16)$$

PHOTO

for different frequencies we use semi classical description

PHOTO upper singe boson lower is fermi we can treat as continuous using integral with single particle density of states bosons special with ground state extra term before integral

keep this term as it indicates if we get BEC

to compute the integral express in terms of polylogarithm

we can use $z = e^{\beta\mu}$. In class we solved for non relativistic particles, we saw cases of relativistic particles.

THIS INTEGRAL IS COMING UP

when computing average energy and eqns of state generally first term doesn't matter. it is crucial if BEC will take place or not.

PHOTO take limits

does BEC occur, easy if N fixed. if N not fixed then we only take average. we have to translate from GC to Canonical. find $\langle N \rangle$ in terms of polylog.

$N = \text{fixed} = \dots \mathcal{L}i_r(z) + N_0(z)$

this is important for C_v

is there a bound in the expression for the polylog most of the time it is bound but for lower maybe not. $0 < z < 1$ monotonic max of $\mathcal{L}i_r(z)$ is at $z = 1$. Critical temp when polylog of $z=1$ $N = N_{ex} = \dots \mathcal{L}i_r(z)$, $\implies T_C$

Fermi temp or fermi energy at $T = 0$

in principle write $N = \int_0^{\infty} \dots$ PHOTO energy has same integral with ϵ term.

For the magnetism part

curie law for paramagnetism $\frac{1}{T}$

we also have negative temperatures see photo left we have alignment with B field, right anti-alignment

For ising model extract info from the transfer matrix